

PART-I (50 points) (Answer ONLY in space provided.)

(Answer all question in the space given; without exceptions. Answers outside this space will NOT be considered; answer in one/two sentences clearly. Weightage is indicated for each question.)

1. Find the terminal velocity of a vertically falling rain drop (mass m) falling from rest and experiencing air resistance; the retarding force is proportional to velocity. Assume that evaporation is negligible and air is still (no turbulence). (2 points)

2. A charged particle (charge q) has velocity $\vec{v}_0$ at $t = t_0$. Show that when magnetic field $\vec{B}(\vec{r}) = B\hat{z}$ is present, the particle undergoes helical motion. What is the pitch of the helical path of this particle? (2 points)

3. Force acting on a gold atom causes its velocity to change from $0.6c$ to $0.8c$. Consider the change in its kinetic energy, in classical mechanics and special theory of relativity. Calculate ratio of changes in KE (relativistic divided by classical). (2 points)

4. A scientific instrument company claims that its newly released microscope can position the proton with an accuracy of $\pm 10^{-11}$ m. Find the uncertainty of the proton's position 1 second later. (2 points)

5. Muon is an unstable elementary particle with mass of $207m_e$; when this muon with a negative charge of $-e$ is 'captured' by proton it forms a 'muonic' atom. Find the ionization energy of this 'muonic' atom in eV. (2 points)

6. When an electron with energy of 1.0 eV is incident on a barrier of 10.0 eV energy and 0.50 nanometer, find (approximately) the transmission probability of this electron. (2 points)

$$T \sim e^{-\sqrt{2 \times 15.32 \times 10^{-9} \times 0.5 \times 10^{-9}} = e^{-0.175}}$$

7. For a harmonic oscillator, find the expectation value of x^2 in $n = 0$ state. (Hint: use Feynman's trick on integral of Gaussian function) (2 points)

8. In Radio astronomy, clouds of hydrogen atoms in galaxy are detected by the 21 cm spectral line; in hydrogen atom, this line corresponds to the flipping of electron spin, from it being parallel to becoming anti-parallel to spin of the proton (nucleus). Find the magnetic field experienced by this electron. (2 points)

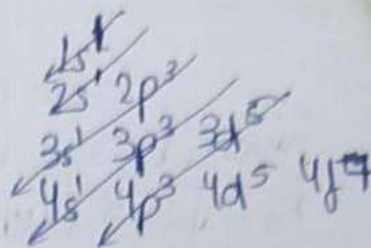
9. What is the physical significance of each quantum number associated with electron in Hydrogen atom? (2 points)

$$E = \mu_B \cdot B$$

[mag. of Ang. mom]

[mag. mom]

10. Phosphorous has atomic number of 15. What will be electronic configuration of this element? (2 points)



11. For an electron in hydrogen atom (excited state), find the value/s of angle between L and z -axis for $l = 2$. (2 points)

(L : Ang. Momentum, m_l : magnetic q. no.)

12. Calculate the probability that a particle in a box L wide can be found between $x = 0$ and $x = L/n$ when it is in n -th state? (2 points)

13. For a particle in a cube of side L , at what temperature is the probability of finding the particle with energy equal to that of the *second* excited state equals one half of its probability in ground state? (Note: find T in terms of ground state energy E_0) (2 points)

14. What is the most probable speed of a oxygen molecule in air at 25 C (assume ideal gas)? Is it faster/slower than an aeroplane (500km/hr)? (2 points)

15. Microprocessors in a computer produces heat at rate of 10 W per square centimeter of surface area. At what temperature would a blackbody be at this radiance? (2 points)

16. From the Plank's radiation law for light intensity $u(\nu)$, find the equation satisfied by peak ν_p (find the most simple algebraic expression that it satisfies). (2 points)

17. The smallest angle of Bragg scattering in potassium chloride (KCl) is 28.4 degrees for 0.30 nm X-Rays. Find the distance between the atomic planes in KCl. (2 points)

18. Otto Cycle is involved in operation of internal combustion engine. The cycle consists of (a) reversible adiabatic compression from A to B , (b) reversible constant-volume pressure increase from B to C caused by combustion of a very small amount of fuel, (c) reversible adiabatic expansion from C to D and (d) reversible constant-volume pressure decrease back to state A . We will denote p_A , V_A , and T_A are pressure, volume and temperature at point A , and likewise for B - D . Answer the following, assuming ideal/perfect gas as the working substance and hence use ideal gas law and the p, V, T relations for adiabats: (Total 7 points)

(a) Draw the p - V diagram for Otto Cycle. (1 point)

(b) What will be the efficiency for this cycle? (1 point)

(c) Derive an expression for the efficiency in terms of V_A and V_B . (2 point)

$$= 1 - \frac{T_c \cdot T_b}{T_d \cdot T_a}$$

- (d) Derive expressions, in terms of $C_{V,m}$ and the temperatures, for the change in entropy (of the systems and surroundings) for each step of the cycle. (2 point)

19. Consider a single particle Lagrangian given by $\mathcal{L} = -m\sqrt{c^2 - \vec{u} \cdot \vec{u}} - V(\vec{r})$ where $\vec{r}$ is the position vector of the particle and has velocity $\vec{u} = \frac{d}{dt}\vec{r}$. And c is a constant. (total 5 points)

- (a) Find conjugate momentum $\vec{p}$ of the particle with Lagrangian $\mathcal{L}$, and hence calculate $\vec{p} \cdot \vec{p}$ in terms of $\vec{u} \cdot \vec{u}$ (2 points)

- (b) For this Lagrangian, find the Hamiltonian. (1)

- (c) Simplify to get a formula for H that looks like the famous Einstein energy-mass relation. (2 points)

20. For Hydrogen atom, it is given that the wavefunction is $\psi(r, \theta, \phi) \propto r^2 e^{-r/3a_0} \sin \theta \cos \theta e^{+2i\phi}$. Calculate to find the quantum numbers associated with this wave function. (4 points)